

OnlineRFPerm with LLM routing — false-alarm rate

Setup. One row is one observation. Y is the numeric outcome, never a feature. X is routed: high-VIMP columns go to a frozen RF, low-VIMP columns go to a TabPFN-style context k NN. The blend is fit once on D_{ref} . Each new batch is $T_t = \text{MSE}_t - E_{\text{ref}}$. ADDIS is the primary OnlineRFPerm mark. The remaining rows are the same MSE stream, other detectors.

Stationary: $Y = X^\top w + \varepsilon$, no onset. Random-noise: X and Y independent $N(0, 1)$, no onset. FAR is the share of replications that reject at least once. There is no labeled change, so any ring is a false alarm.

Table 1: False-alarm rate on streams with no labeled onset. Routing is high-VIMP RF + low-VIMP TabPFN-style kNN. FAR is the share of 25 replications that reject at least once over 20 batches. ADDIS is the primary OnlineRFPerm mark.

method	stationary FAR (%)	random-noise FAR (%)
OnlineRFPerm + ADDIS	0.0	0.0
OnlineRFPerm + SAFFRON	0.0	0.0
OnlineRFPerm + fix- α	60.0	80.0
last-two hop ($1.5\times$)	96.0	92.0
Page–Hinkley	64.0	40.0
EWMA	8.0	0.0
CUSUM	48.0	32.0
DDM	4.0	4.0
ADWIN-lite	4.0	4.0
betting martingale	20.0	36.0